

Project Design Phase-II

Data Flow Diagram & User Stories

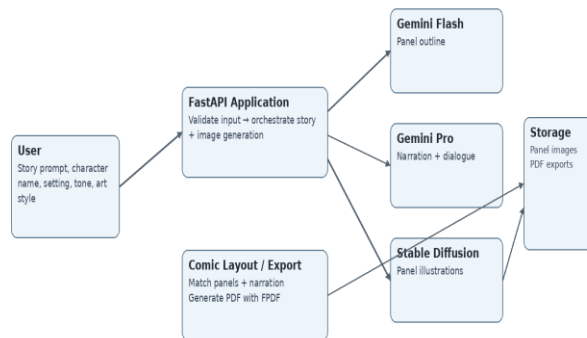
Date	30 September 2026
Team ID	SWTID-2026-1429
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [\(Simplified\)](#)

ComicCraft – Level 0 Data Flow Diagram



Main flow: User Input → FastAPI → AI generation → Layout → PDF/Files → User download.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Web user)	Story Input & Customization	USN-1	As a user, I can enter a story prompt, character name, setting, tone and art style.	The form accepts all required inputs and submits them to the FastAPI /generate route.	High	Sprint-1
Customer (Web user)	AI Story Generation	USN-2	As a user, I can generate a structured comic outline and detailed story from my prompt.	A 5-panel outline, narration and character dialogue are generated.	High	Sprint-1
Customer (Web user)	AI Illustration	USN-3	As a user, I can receive an illustration for each generated comic panel.	Each panel has a corresponding AI-generated comic-style image.	High	Sprint-1
Customer (Web user)	Comic Preview	USN-4	As a user, I can preview my complete comic panel-by-panel before export.	The preview displays panel title, image, scene description, captions and narration.	High	Sprint-1
Customer (Web user)	PDF Export	USN-5	As a user, I can download my generated comic as a PDF.	A multi-page PDF is generated and the export success page confirms completion.	High	Sprint-1
Developer / API Client	JSON Generation API	USN-6	As an API client, I can submit a JSON request to generate a comic.	/generate-comic/json returns layout data and the generated PDF path.	Medium	Sprint-2
Developer / Tester	Image Testing	USN-7	As a developer, I can test image generation independently with a prompt.	/test-image returns a generated image path for the supplied prompt.	Medium	Sprint-2
Administrator / Developer	Configuration	USN-8	As a developer, I can keep Gemini and Hugging Face credentials outside source code.	API keys/tokens are read from environment variables or a .env configuration.	High	Sprint-1
Administrator						

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